

ARTICLE

Can large language models provide accurate and empathetic answers to the most frequently asked questions by infertile patients? A pilot study



BIOGRAPHY

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KEY MESSAGE

Large language models can provide answers that are as good and as empathetic as those provided by professionals to questions frequently asked by patients in assisted reproductive technology centres. Nevertheless, careful oversight and regulation are essential to ensure the accuracy and appropriateness of the information provided.

ABSTRACT

Research question: Is the quality, relevance and empathy of the answers provided by large language models (LLMs) in response to the most frequently asked patient questions in reproductive medicine comparable to those provided by human specialists?

Design: This monocentric, double blind, prospective study involved two clinicians and two embryologists who answered 13 frequently asked questions in their respective field. The same questions were asked to a free online LLM, with the same constraint of text length as practitioners. All answers were blindly evaluated by four assessors (two gynaecologists and two embryologists depending on the topic) for quality and accuracy. A psychologist also evaluated empathy.

Results: The mean number of words per answer was significantly higher ($P < 0.001$) for LLM than for humans. The average quality of answers was not statistically different between LLM and professionals. No answer provided by LLM was evaluated as completely aberrant, and only a minority contained false or inappropriate information or was scored as being very poor by assessors. Answers provided by embryologists, but not clinicians, ranked significantly higher ($P = 0.02$) than LLM. The psychologist chose LLM answers as most empathetic, clear, or both, in 14 out of 26 questions.

Conclusions: LLMs could be used as an educational tool within assisted reproductive technology centres to answer frequently asked patient questions. Although the potential applications of LLMs' capabilities in answering medical questions are numerous, this should be carefully evaluated and regulated to prevent the dissemination of inaccurate information to patients.

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KEYWORDS

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Frequently asked questions
Infertility patients

INTRODUCTION

In recent years, the rapid advancement of artificial intelligence (AI) has notably transformed various medical fields, including assisted reproductive technology (ART). Artificial intelligence now offers innovative tools for assessing embryo quality, selecting spermatozoa and managing ovarian stimulation, ultimately holding promise for improving overall ART success rates. Beyond these core applications, other AI-driven technologies are also becoming integral to the fertility journeys of patients. Notably, large language models (LLMs) have gained traction as accessible, real-time sources of information for individuals seeking guidance on reproductive health and fertility treatments.

These AI-based platforms draw upon vast datasets, including published research and patient experiences, to provide comprehensive answers. Their appeal lies in their flexibility, availability and immediacy, especially in busy ART centres in which healthcare professionals often have limited time to respond to the growing number of patient inquiries. Concerns persist, however, regarding the reliability, accuracy and empathetic quality of AI-generated responses.

The expanding use of AI chatbots and virtual assistants in health care raises important ethical and practical questions. Recent studies suggest that AI can improve access to medical knowledge by offering instant answers to common questions about fertility, ovulation, IVF success rates and treatment options (Olawade *et al.*, 2025). Despite these advantages, AI responses may lack nuance, misinterpret complex cases or fail to deliver personalized advice. Moreover, conveying empathy (an essential component of effective patient-provider communication) remains an important challenge for AI (Inzlicht *et al.*, 2024; Roshanaei *et al.*, 2024; Aravazhi *et al.*, 2025).

Comparative studies in other medical fields, such as dermatology, have demonstrated high diagnostic accuracy from AI models but limited capacity for holistic, patient-centred guidance (Aravazhi *et al.*, 2025; Busch *et al.*, 2025). In reproductive medicine, emotional support is as crucial as clinical accuracy; however, the effectiveness of AI in addressing patient concerns is still largely unexplored. A recent study evaluating the

ability of LLMs to predict pregnancy outcomes after IVF treatment represents one of the few exceptions (Cao *et al.*, 2025).

Given the psychological burden associated with infertility treatments (Boivin *et al.*, 2021), it is vital to determine whether AI platforms can offer factually accurate information as well as reassurance and empathetic communication. Understanding the evolving role of AI in reproductive medicine is, therefore, essential for shaping best practices that integrate AI-assisted consultations with specialist-led care.

The aim of the present study was to evaluate and compare the quality, relevance and empathy of responses provided by LLMs versus those given by human specialists in response to the most frequently asked patient questions in reproductive medicine.

MATERIALS AND METHODS

This monocentric prospective study was conducted in a university-based hospital in France. No patient data were used; therefore, no specific ethical approval was requested per current regulation. Two gynaecologists and two embryologists were invited to assess responses. The questionnaire included 26 of the most frequently asked questions in the field of assisted reproduction according to experience, both clinically ($n = 13$) and biologically ($n = 13$). They were also submitted to an online free LLM (ChatGPT v4.0). The only instruction given to respondents and LLM was to provide a written answer (in French) that was a maximum of five lines long.

After anonymization, all answers were blindly evaluated by assessors, either two gynaecologists or two embryologists depending on the topic, according to the following scale: is the answer aberrant (yes/no)? Does the answer contain any false or inappropriate information? Is the answer consistent with current French regulation, clinical guidelines (not at all/partly/yes/not applicable), or both? What is your overall evaluation of the answer (very poor/poor/acceptable/fair/very good)? For this last item, the answer was converted to numbers for further statistical analysis (very poor = 1 point up to very good = 5 points). For each item, assessors also had to rank answers from best (1) to poorest (3). Finally,

a psychologist evaluated all answers and ranked them according to empathy and clarity for each item.

Graphpad Prism software version 9 (GraphPad Software, Boston, Massachusetts USA) was used for all statistical analysis (Student's *t*-test or Mann–Whitney tests were used where appropriate). $P < 0.05$ was considered statistically significant.

RESULTS

Although the instruction on the length of answers was the same for all respondents (LLM and humans), the mean number of words per answer was significantly higher for LLM than for humans (64.7 ± 14.1 versus 44.5 ± 16.4 , $P < 0.001$). No significant difference was observed across human participants (data not shown).

The distribution of the quality of the answers given by LLM or ART professionals (gynaecologists and embryologists) is presented in [FIGURE 1A](#). After arbitrarily conversion into numbers (1 = very poor/5 = very good), the mean quality of answers was not statistically different between LLM and professionals (3.65 ± 0.72 versus 3.66 ± 0.55 , $P = 0.95$).

When the 13 questions related to embryology and biological aspects of ART were considered, no answer was evaluated as completely aberrant, but three out of 13 answers provided by LLM were considered as containing false or inappropriate information and were categorized as poor or acceptable. Most answers were evaluated as acceptable or fair. The distribution of answer quality given by LLM or embryologists is presented in [FIGURE 1B](#). After conversion into numbers (1 = very poor/5 = very good), the mean quality of answers was not statistically different between LLM and embryologists (3.335 ± 0.55 versus 3.67 ± 0.40 , $P = 0.1$). Answers provided by embryologists, however, ranked significantly higher than LLM (1.79 ± 0.3 versus 2.46 ± 0.6 , respectively, $P = 0.02$).

When subgroup analysis was conducted in the 13 questions related to clinical and gynaecological aspects of ART, no answer was evaluated as aberrant or containing false or inappropriate information or being very poor by assessors (gynaecologists [IVF specialists]). Most answers were evaluated as fair or very good. The distribution of the

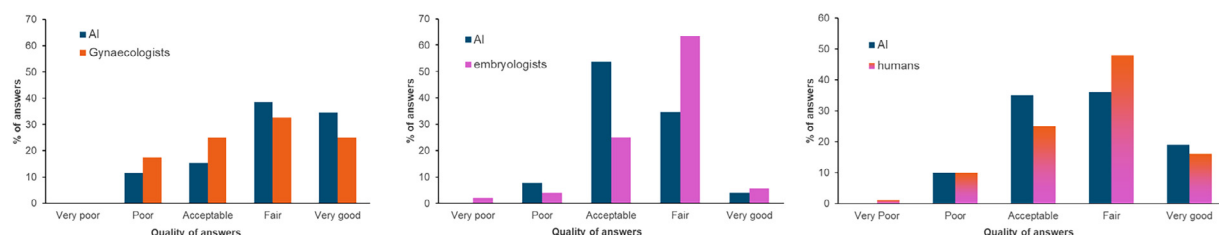


FIGURE 1 (A) Distribution of the quality of the answers given by large language models (LLM) versus assisted reproductive technology professionals (gynaecologists and embryologists), (B) by LLM versus embryologists and (C) by LLM versus gynaecologists.

quality of the answers given by LLM or gynaecologists is presented in [FIGURE 1C](#). After conversion into numbers, the mean quality of answers was not statistically different between LLM and gynaecologists (3.96 ± 0.75 versus 3.65 ± 0.65 , $P = 0.3$). When ranking of answers is considered (1 = best answer/3 = poorest answer), no statistical difference was observed either between LLM and gynaecologists (1.65 ± 0.8 versus 2.17 ± 0.4 , $P = 0.05$).

The psychologist chose LLM answers as most empathetic, clear, or both, in 14 out of 26 questions ($n = 8$ most empathetic and $n = 6$ clearest answer).

DISCUSSION

In the present study, we compared responses to frequently asked patient questions provided either by a publicly available LLM or by ART professionals, including gynaecologists and embryologists. Our findings show that the LLM did not underperform compared with human professionals in response quality and empathy, and, in some cases, it even outperformed them.

In recent years, LLMs, also known as conversational agents, have made remarkable progress, capturing the interest of the public and healthcare professionals. Numerous studies have evaluated their ability to answer medical questions across various specialties, including pulmonology ([Bernstein et al., 2023](#)), orthopaedic surgery ([Zhang et al., 2024](#)) and ophthalmology ([Tailor et al., 2024](#)). A broader study involving 33 clinicians across 17 specialties also assessed LLM performance across diverse domains ([Goodman et al., 2023](#)). Collectively, these investigations found that LLMs generally provide satisfactory answers to common medical questions. Limitations, however, were also highlighted. Although LLMs often offer accurate responses to basic queries,

experts have identified instances of incomplete or even irrelevant answers, underscoring the need for human oversight when using AI in medicine ([Au Yeung et al., 2023](#); [Ullah et al., 2024](#)). Studies comparing LLM outputs with those from healthcare professionals across clinical contexts have reached similar conclusions ([Ayers et al., 2023](#); [Bernstein et al., 2023](#); [Maida et al., 2024](#); [Ye et al., 2024](#)). These works demonstrate that the accuracy of AI-generated responses varies considerably, depending on the complexity and specificity of the question ([Avnat et al., 2024](#); [Lievin et al., 2024](#)). Although LLMs tend to excel with general inquiries or straightforward cases, their performance declines when faced with more complex scenarios that require contextual judgment. In contrast, healthcare professionals offer more accurate and personalized responses tailored to individual patient needs.

One particularly noteworthy aspect of conversational agents is their ability to generate responses that may be perceived as empathetic. Some studies ([Ayers et al., 2023](#); [Maida et al., 2024](#)) have shown that LLM-generated replies are often longer and more detailed, and are easier for patients to understand, especially on general health topics. The compassionate tone and phrasing used by LLMs can, in some cases, make their responses seem more empathetic than those from human professionals. This, however, raises a critical question: are these responses truly appropriate in medical contexts in which individual patient nuance is essential? Some researchers have warned that perceived empathy may obscure inaccuracies or incompleteness in the information provided, particularly in more complex situations. To explore this issue further, two key questions arise. First, should we use LLMs in contexts requiring empathetic care, regardless of user preferences? We argue that this is ethically problematic. As highlighted in a recent

correspondence, LLMs simulate, but do not experience, emotions. Relying on them for empathetic care risks deception, especially if users mistake simulated concern for genuine understanding. This may undermine human connection and cause emotional harm. Similarly, [Inzlicht et al. \(2024\)](#) acknowledged the potential benefits of AI-simulated empathy but cautioned against using it without transparency, as this could exploit user vulnerability and lead to harmful overdependence. Ethical use in such contexts, therefore, requires user awareness and consent. Second, should user preferences shape the ethical assessment of medical technologies like LLMs? Although both studies ([Inzlicht et al., 2024](#)) note that users may feel comforted or emotionally connected to empathetic AI, ethical evaluation must go beyond immediate preferences. Apparent satisfaction may be rooted in self-deception rather than informed consent. A robust ethical approach must consider long-term effects, such as the preservation of authentic human relationships and the risk of diminishing human empathy.

The findings from the present study on an LLM's ability to respond to common ART-related questions align with previous research. Overall, the quality of responses did not significantly differ between the LLM and human specialists ([Wang et al., 2024](#)). The present study, however, identified marginal, non-significant trends: although evaluators generally preferred LLM-generated responses over those from gynaecologists, they tended to favour embryologists' answers for biologically oriented questions. In fact, some of the LLM's answers to biological questions contained inaccuracies and were rated lower in quality, although none were classified as 'very poor'. Whether the accuracy of LLMs varies by medical subspecialty remains an underexplored area ([Goodman et al., 2023](#)) and warrants further investigation.

A novel aspect of our study was the inclusion of a psychologist to assess the empathy of responses provided by both the LLM and human professionals. This analysis found that empathy levels were comparable across both groups. Interestingly, the LLM's higher word count did not influence the empathy evaluation. Nonetheless, other studies have reported higher empathy and perceived usefulness scores for LLMs when rated by patients (Armbruster *et al.*, 2024). This raises important methodological questions: who should assess AI-generated empathy, and what tools or frameworks should be used? Future studies should consider using standardized empathy rating scales or semi-structured interviews to address this question more robustly.

The present study has several strengths. First, it included evaluators from diverse disciplines (gynaecologists, embryologists and a psychologist) ensuring a multi-faceted assessment. Second, both informational accuracy and the empathetic tone of responses were examined. There are notable limitations, however. Being a monocentric study, the results may reflect institutional biases or shared viewpoints among respondents and evaluators. Expanding the study across multiple centres and involving external evaluators could help minimize these biases and improve generalizability. Such multicentre research should include a standardized evaluation protocol and training for assessors. The limited sample size also restricts the statistical power of our findings, and results should be interpreted with caution. Additionally, despite anonymization, evaluators may have been able to identify AI-generated responses, potentially influencing their assessments. Interestingly, some evaluators suggested that an 'ideal' answer might combine the strengths of human and LLM-generated responses, a concept recently referred to as 'expert-edited LLMs' (Tailor *et al.*, 2025). This suggests that hybrid tools blending AI efficiency with human clinical judgment may represent a promising direction. Another consideration is the language used by the LLM. Most LLM training data are based on English-language sources, giving the models a more nuanced understanding of English grammar, terminology and cultural context. This could lead to reduced accuracy in other languages, such as French, which was used in the present study. As such, our findings may not fully

apply to LLM performance in English or other languages. Future studies should evaluate LLMs' multilingual capabilities more systematically. Finally, the question of how empathy is evaluated remains open. Although our psychologist evaluator applied consistent criteria, focusing on warmth and clarity, future research would benefit from the use of validated empathy measurement tools or standardized input prompts.

In conclusion, this pilot study demonstrates that LLMs may serve as useful educational tools within ART centres for addressing patients' frequently asked questions on general topics. Centralizing these AI-generated responses on platforms such as clinic websites could offer patients easy access to reliable information while reducing the burden on healthcare teams to repeatedly answer routine inquiries. Nevertheless, the integration of LLMs into patient care must be approached with caution. Although their potential is substantial, careful oversight and regulation are essential to ensure the accuracy and appropriateness of the information provided.

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